STAT 110

Lecture 1: Probability & Counting

- A **sample space** is the set of all possible outcomes of an experiment.
- An **event** is a subset of a sample space.
- The probability of an event A is

$$P(A) = \frac{\text{number of favorable outcomes to } A}{\text{number of possible outcomes}}.$$

- **Example:** Flip a fair coin **twice**. The sample space is

$$S = \{HH, HT, TH, TT\}.$$

If event A is "both flips are tails," then

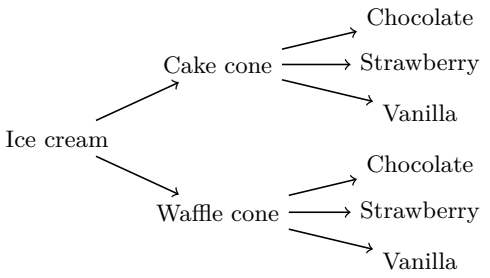
$$A = \{TT\}, \qquad P(A) = \frac{1}{4}.$$

- A **fair event** assumes all outcomes are equally likely and the sample space is finite.

Counting

Multiplication Rule. If the first experiment has n_1 possible outcomes, and for each outcome of the first experiment there are n_2 outcomes for the second experiment, and so on through the rth experiment, then the number of possible overall outcomes is

$$n_1 n_2 \cdots n_r.$$



$2 \cdot 3 = 6$ possible outcomes

Example: Full House in Poker (5-card hand)

A deck has 52 cards. We receive a 5-card hand from a shuffled deck, so all hands are equally likely.

Question: What is the number of possible hands? There are $\binom{52}{5}$.

Answer (probability of a full house):

$$P(\text{full house}) = \frac{13 \times \binom{4}{3} \times 12 \times \binom{4}{2}}{\binom{52}{5}}.$$

Binomial coefficient:

$$\binom{n}{k} = \frac{n!}{(n-k)!\,k!} \quad (0 \le k \le n), \qquad \binom{n}{k} = 0 \quad \text{if } k > n.$$

This counts the ways to choose k people from n people, i.e., a subset of size k where **order does not matter**.

Sampling Table: Choose k Objects from n

1. Replacement: we can pick the same person again.
2. No replacement.
3. Order matters.
4. Order does not matter.

	Order matters (3)	**Order does not matter (4)**
Replacement (1)	n^k	$\binom{n+k-1}{k}$
No replacement (2)	$n(n-1)\cdots(n-k+1)$	$\binom{n}{k}$